Use an Adaline network to train ANDNOT function with bipolar inputs and targets. Perform two epochs of training

ANDNOT

x_1	x_2	1	t
1	1	1	-1
1	-1	1	1
-1	1	1	-1
-1	-1	1	-1

$w_1 = w_2 = b = 0.2$

$\alpha = 0.2$

[1, 1, -1]

$y_{in} = b + x_1 w_1 + x_2 w_2 = 0.2 + 1 \times 0.2 + 1 \times 0.2 = 0.6$

$(t - y_{in}) = -1 - 0.6 = -1.6$

$w_1(new) = w_1(old) + \alpha(t - y_{in}) x_1 = 0.2 + 0.2 \times (-1.6) \times 1 = -0.12$

$w_2(new) = w_2(old) + \alpha(t - y_{in}) x_2 = 0.2 + 0.2 \times (-1.6) \times 1 = -0.12$

$b(new) = b(old) + \alpha(t - y_{in}) = 0.2 + 0.2 \times (-1.6) = -0.12$

$E = (t - y_{in})^2 = (-1.6)^2 = 2.56$

[1 -1 1]

$y_{in} = b + x_1 w_1 + x_2 w_2 = -0.12 + 1 \times -0.12 + 1 \times -0.12 = -0.12$

$(t - y_{in}) = 1 - -0.12 = 1.12$

$w_1(new) = -0.12 + 0.2 \times (1.12) \times 1 = 0.10$

$w_2(new) = -0.12 + 0.2 \times (1.12) \times -1 = -.34$

$b(new) = -0.12 + 0.2 \times (1.12) = 0.16$

$E = (1.12)^2 = 1.25$

[-1 1 -1]

$y_{in} = b + x_1 w_1 + x_2 w_2 = 0.10 + -1 \times 0.10 + 1 \times -.34 = -0.34$

$(t - y_{in}) = 1 - -0.34 = -0.66$

$w_1(new) = 0.10 + 0.2 \times (-0.66) \times -1 = 0.24$

$w_2(new) = -0.34 + 0.2 \times (-0.66) \times 1 = -0.48$

$b(new) = 0.10 + 0.2 \times (-0.66) = -0.03$

$E = (-0.66) = 0.43$

[-1 -1 -1]

$y_{in} = -0.03 + -1 \times 0.24 + -1 \times -0.48 = 0.21$

$t - y_{in} = 1 - 0.21 = 1.2$

$w_1(new) = w_1(old) + \alpha(t - y_{in}) x_1 = 0.24 + 0.2(1.2) \times -1 = 0.48$

$w_2(new) = -0.48 + 0.2(1.2) \times -1 = -0.23$

$b(new) = -0.03 + 0.2(1.2) = 0.21$

$E = (t - y_{in})^2 = (1.2)^2 = 1.44$

EPOCH-2

[1, 1, -1]

$y_{in} = 0.21 + 1 \times 0.48 + 1 \times 0.23 = 0.92$

$t - y_{in} = 1 - 0.92 = 0.98$

$w_1(new) = 0.48 + 0.2(0.98) \times 1 = 0.68$

$w_2(new) = -0.23 + 0.2(0.98) \times 1 = -0.43$

$b(new) = 0.21 + 0.2(0.98) = 0.46$

$E = (0.98)^2 = 0.95$

[1, -1, 1]

$y_{in} = 0.46 + 1 \times 0.68 + -1 \times -0.43 = 0.25$

$t - y_{in} = 1 - 0.25 = 0.76$

$w_1(new) = 0.68 + 0.2(0.76) \times 1 = 0.83$

$w_2(new) = -0.43 + 0.2(-0.43) \times -1 = -0.56$

$b(new) = 0.46 + 0.2(0.46) \times 1 = 0.31$

$E = (0.76)^2 = 0.57$

[-1 1 -1]

$y_{in} = 0.31 + -1 \times 0.83 + 1 \times -0.56 = -1.33$

$t - y_{in} = 1 - -1.33 = 0.33$

$w_1(new) = 0.83 + 0.2(0.33) \times 1 = 0.97$

$w_2(new) = -0.56 + 0.2(0.33) \times 1 = -0.51$

$b(old) = 0.31 + 0.2(0.33) \times -1 = 0.25$

$E = (0.33)^2 = 0.106$

$-1 \quad -1]$

$y_{in} = -0.25 + -1 \times 0.37 + -1 \times -0.51 = -0.11$

$t - y_{in} = (1 - 0.11) = .90$

$w_1(new) = 0.37 + (0.2 \times 0.90 \times -1) = 0.55$

$w_2(new) = -0.51 + 0.2(-0.90) = -0.38$

$b(new) = -0.25 + 0.2(-0.90) = 0.43$

$E = (.90) = 0.8$

Input						weights			Error
x_1	x_2	1	t	y_{in}	$(t-y_{in})$	w_1	w_2	b	
EPOCH-1									
1	1	1	-1	0.6	-1.6	-0.12	-0.12	-0.12	2.56
1	-1	1	1	-0.12	1.12	0.10	-0.34	0.16	1.25
-1	1	1	-1	-0.34	-0.66	0.24	-0.48	-0.03	0.43
-1	-1	1	-1	0.25	-1.2	0.48	-0.23	0.27	1.47
EPOCH-2									
1	1	1	-1	-0.00	-0.98	0.28	0.43	-0.46	0.95
1	-1	1	-1	0.25	0.76	0.43	-0.58	-0.31	0.57
-1	1	1	-1	-1.33	0.33	0.37	-0.51	-0.25	0.106
-1	-1	1	-1	-0.11	-0.55	0.55	-0.38	0.43	0.8

12b) State the concept of the Delta rule used in ADALINE

The Delta Rule is used to train ADALINE

Key Concept of the Delta Rule

1. Based on Minimization of Mean Square Error

The rule adjusts weights to minimize the square error b/w the target o/p and actual o/p

Error calculation

$E = t - y_{in}$

t = target o/p

y = actual o/p

Weight Update Formula

$w_{new} = w_{old} + \alpha e x$

α = learning rate

e = error

x = input

Bias update

$b_{new} = b_{old} + \alpha e$

Uses Linear Activation function

Unlike perception learning, ADALINE updates weights using the net i/p

Gradient Descent Method

The delta rule follows the steepest descent method to reduce error gradually